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324356(24)

B. E. (Third Semester) Examination, April-May 2015

(New Scheme)

(Elect. Engg.)

ELECTRICAL POWER GENERATION

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Part (a) of each question is compulsory and carry 2 marks. Attempt any two from part (b), (c) and (d). Each carry 7 marks.*

Unit-I

1. (a) Define plant capacity factor & how it is related to lead factor?

(b) What is the difference between conventional and non-conventional energy sources? What is the need for non-conventional source.

(c) Explain diversity factor. State the advantages of

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diversity of load in a power supply system. Show the load factor of power plant is improved by increase in diversity of load.

(d) A power system had the daily load curve given following table :

Time	Load in MW
12.00 night to 2 am	20
2 am to 8 am	10
8 am to 12:30 noon	50
12:30 noon to 1:00 pm	40
1:00 pm to 6 pm	50
6 pm to 12 night	70

Plot the following curve :

(a) Chronological load curve

(b) Load - duration curve

(c) Load - energy curve

Calculate the load factor and utilization factor of plant if the installed capacity is 100 MW.

Unit-II

2. (a) Define saturated and superheated steam.

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(b) Describe different types of steam turbines.

(c) Explain with neat sketch the functions of the following in a steam power station :

(i) Economises

(ii) Air preheater

(iii) Boiler

(iv) Condenser

(d) Draw the layout of a steam power plant & show all parts with their brief function.

Unit-III

3. (a) What is nuclear fission?

(b) Explain hydrograph and its use in the study of water power. Also explain how to construct a flow duration curve from a hydrograph.

(c) Explain the following with respect to hydro power plants :

(i) Surge tank

(ii) Penstock of tailrace

(d) What are the different types of reaction? Briefly state their functions.

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Unit-IV

4. (a) Define photovoltaic cell.

(b) Describe various types of solar collector. How is solar energy converted to electrical energy.

(c) Describe with neat sketch the working of a wind energy system with main components.

(d) Discuss the advantages and disadvantages of wind energy conversion system.

Unit-V

5. (a) What is base load and peak laod?

(b) Discuss closed cycle MHD system with neat sketch.

(c) What are the main applications of geothermal energy, Biomass and ocean thermal energy?

(d) Explain tidal power generation.

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